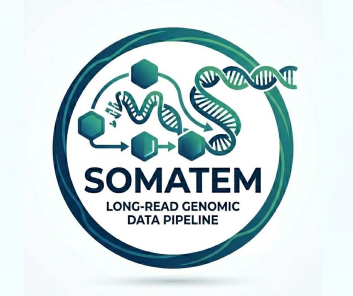
Treangen lab
@ Rice University

RICE ENGINEERING AND COMPUTING
Department of Computer Science



somatem: an open source and modular analysis tool for ensemble long-read metagenomics

Prashant Kalvapalle, Austin Marshall, Dongwei Li, Todd Treangen, *et. al.*

Mid-atlantic microbiome meetup, 20/Mar/26

The learning curve: my entry into metagenomics

- Synthetic biologist: engineering plasmids -> wastewater microbiology
- Task: Identify species

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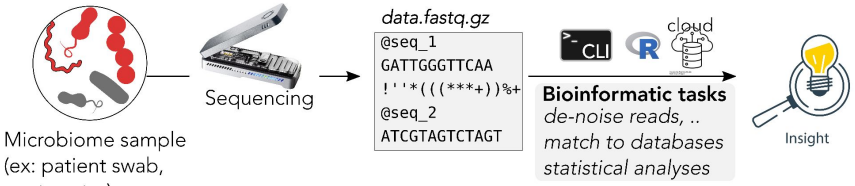
Long read, shotgun seq

- Many software
- Fewer pipelines



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- Need to find and pick a tool, database
- Configure thresholds, other settings
- Any caveats in data interpretation, ex: false positives

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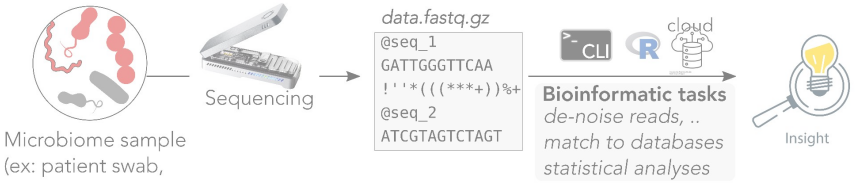

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Our solution:

- `somatem`/CLI : modular metagenomics pipeline
- `omi`/GUI : Co-pilot to [*configure, run and explore*] pipeline results

Context: somatem evolved from MetAMOS

Short-reads

Home > Genome Biology > Article

MetAMOS: a modular and open source metagenomic assembly and analysis pipeline

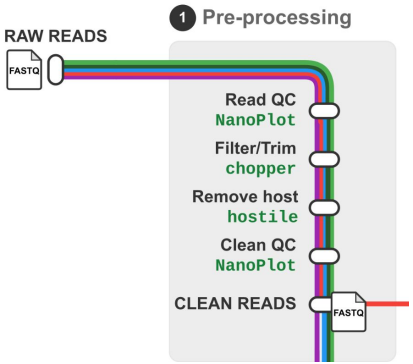
Software | Open access | Published: 15 January 2013

Long-reads

	mmlong2	somatem
Orchestrator	Snakemake	Nextflow
End goal	Assembly	1) Assembly 2) Tax profiling 3) SV detection 4) Pathogen detection
Co-pilot	-	omi

somatem: pipeline for long-read metagenomics

Dr. Austin Marshall



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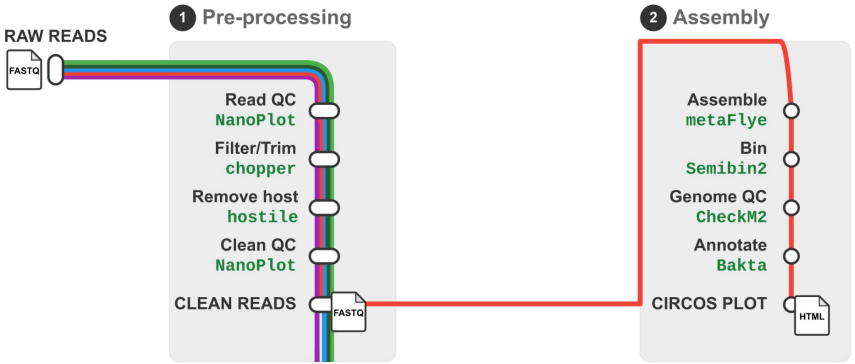

Assemble genomes

somatem: pipeline for long-read metagenomics

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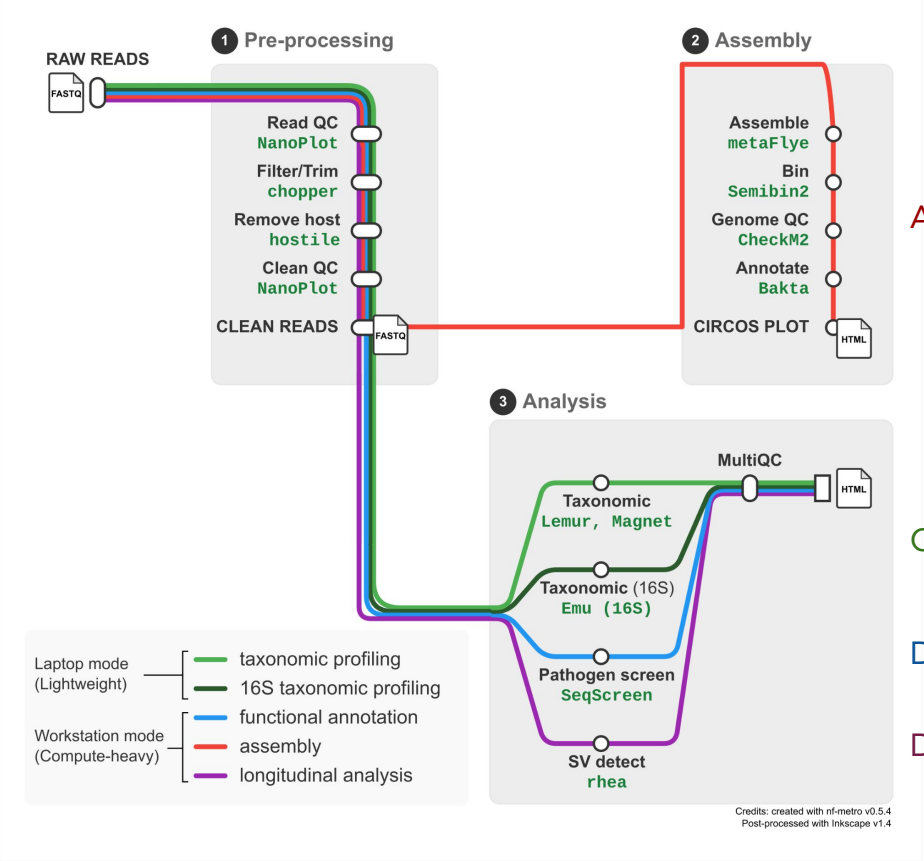

Assemble genomes

Classify species, profile

Detect virulence/resistance genes

Discover structural variants in time-series

Chart: https://github.com/pinin4fjords/nf-metro

somatem: easy setup, modularity and reproducibility

- Ease of setup
  - Single install: GitHub CONDA
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    - (*..in progress*)
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  - Self contained run information in plain text configuration files

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      input: "assets/meta_tax_samplesheet.csv"
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Run with: `nextflow run somatem.nf -config *.yaml`

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- Documentation
  - Wiki: walk-through examples
  - Test datasets downloaded

Appropriate analysis still requires some knowledge

- Articulation: bioinformatics-speak
  - *"Identify organisms"* -> "`taxonomic profiling`"
- Knowledge: parameters to change?
  - High stringency: remove low abundance
- Interpretation: How do I understand these outputs?
  - Why can't I do regular stats with relative abundances

Introducing Omi: Your Metagenomics Co-pilot

Dongwei Li

- LLM with Bioinformatic knowledge + runs pipeline remotely

User query
Analysis goal

→

RAG Database
Bioinformatics papers

→

LLM interface
API or local use

→

Run pipeline
cloud / local server

→

Pipeline results
Interactive; explain..?

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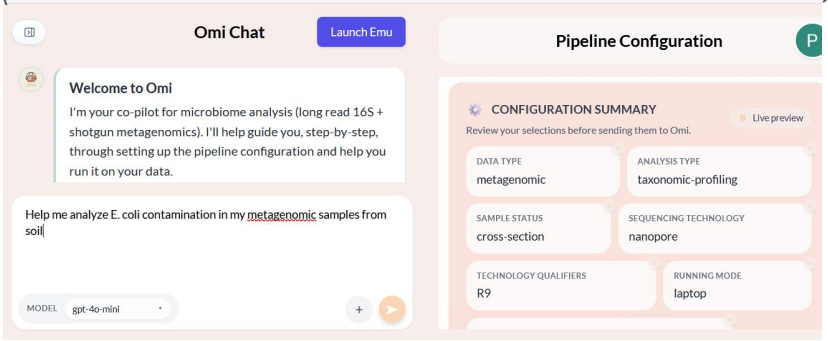
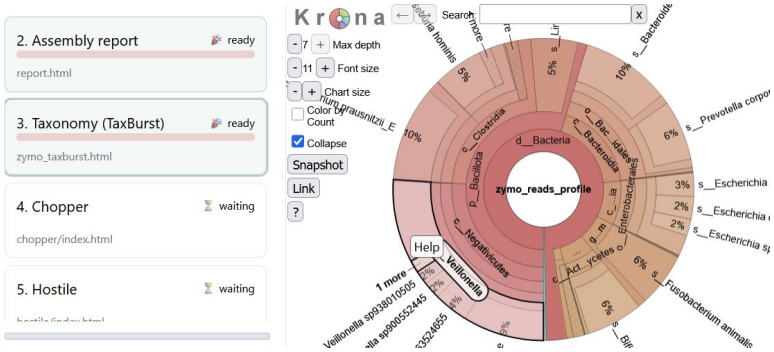
Remote web app: `omi.rice.edu/chat`



Live results



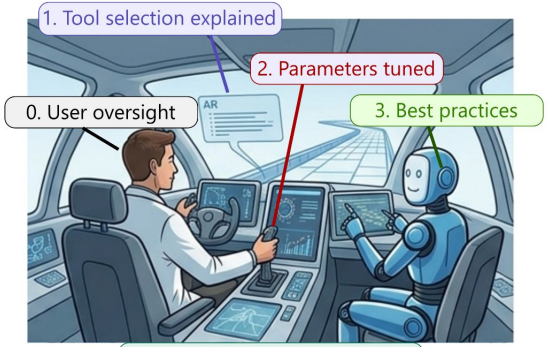
Omi is not a push-button agent!

Push button agent

Results appear...
but how? why?

? Still confused

Omi Co-pilot



Results + Understanding
Reproducible workflow

✓ In control & learning

- Easier learning curve
  - ↓ *pain/friction*
  - But stay in control
- Educational tool: builds bioinformatics literacy
- Ensures best practices pipelines ; reproducibility

Illustration co-generated with Gemini, Claude sonnet 4.5; composed in Inkscape

Acknowledgements

Treangen lab

Authors: Prashant Kalvapalle, Austin G. Marshall, Dongwei Li, Todd J. Treangen
Sahil Joshi, Benjamin Mao, Marko Tanevski, Eddie Kim, Anshumali Shrivastava

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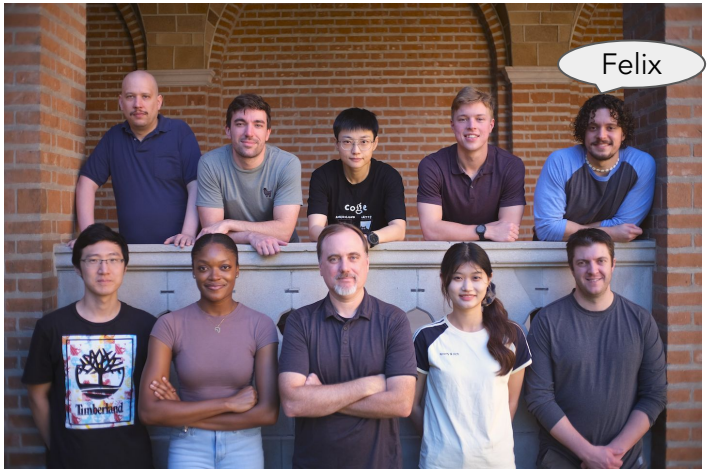

Rush lab

Dr. Anshumali Srivastava

Sahil Joshi

https://treangenlab.com/

Funding sources

RICE ENGINEERING AND COMPUTING
Department of Computer Science

RICE UNIVERSITY
Ken Kennedy Institute
Responsible AI and Computing for Global Impact

AI2Health & GenAI clusters

Somatem democratises microbiome discovery

1. Best-practices nextflow pipeline: *long-read metagenomics*
2. By "collaborating" with a domain specific LLM-co-pilot

SOMATEM
LONG-READ GENOMIC DATA PIPELINE

Experimental biologist

- Determine questions
- Experimental setup, processing
- Data collection

Feedback

Co-pilot: `omi.rice.edu`

- Identify suitable tools
- Educate about caveats
- Appropriate parameters
  - *Ex: k-mer length*
- Interpreting results (*future*)
  - Statistical analysis
  - Data visualization

Bioinformatics

data.fastq.gz

```
@seq_1
GATTGGGTTCAA
!''*(((***+))%+
@seq_2
ATCGTAGTCTAGT
```

CLI cloud

Bioinformatic tasks
de-noise reads, ..
match to databases
statistical analyses

Insight

GitHub: treangenlab/somatem